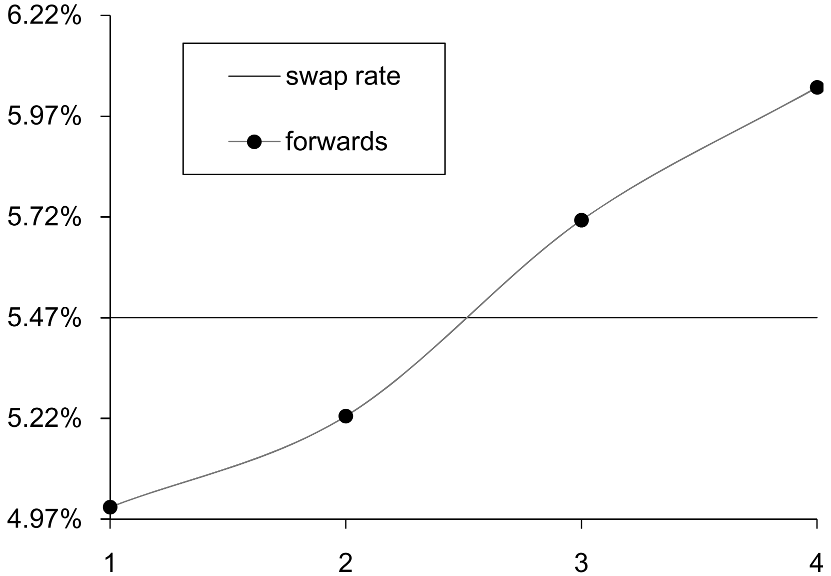

Figure 3.7: The Relationship Between the 4-Year Swap Rate and the Forwards

payment at time t. The 4-year swap rate, s_4, that we solved for is equal to

$$s_4 = \sum_{t=1}^{4} w_t \times f(t-1, t)$$

and is therefore a weighted average of the forward rates.

Figure 3.7 also demonstrates that in the case of an upward-sloping forward curve, the swap rate of a given maturity is greater than the initial forward rate, which is of course today's LIBOR setting. This is to say that the fixed rate receiver in this swap will earn positive cash flow in the initial period. This is commonly referred to as *positive carry*. Thus, this swap has no cash flows when it is initiated (it is an NPV = 0 transaction) and will have positive cash flow for the fixed rate recipient at the end of the first year. Of course, this in no way means that there is an arbitrage. According to the forwards, the fixed rate receiver is expected to earn positive carry at the beginning of the trade and negative carry (or negative cash flow) after some point. The fixed rate in the swap is such that the present value of the positive carry that is expected exactly offsets the present value of the negative carry.

It is important to remember that these forwards may or may not (and

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	5,466,566	5,206,253	5,000,000	4,761,905
2	5.113%	5.226%	5,466,566	4,947,699	5,225,748	4,729,739
3	5.312%	5.712%	5,466,566	4,680,348	5,712,202	4,890,656
4	5.494%	6.041%	5,481,543	4,425,792	6,041,364	4,877,792
				19,260,092		19,260,092
		Fixed rate	5.392%	NPV	0	
		Notional	100,000,000			

Figure 3.8: The Swap Model Assuming the Fixed Side Pays Annual Money

they usually won't) come to fruition. It is only the first LIBOR setting in a spot starting swap that is known with certainty, and all subsequent LIBOR settings in the swap will not be known until they occur. Inasmuch as the actual LIBOR settings differ from the rates implied by the forwards, one may have either positive or negative profits from holding on to the swap until maturity.

A Change in Day Count

By the way, what is the swap rate if we assume the fixed side pays annual money instead of annual bond (and assume the floating side still pays 1-year LIBOR annual bond)? If we use Equation 1.9, given a swap rate of 5.47% annual bond, we see that the annual money fixed rate should be approximately

$$5.47\% \times 360/365 = 5.395\%. \tag{3.8}$$

But in Section 1.2.1 we emphasized that the only way to answer a question like this precisely is to use a swap model. Refer now to Figure 3.8, which shows the swap modeled up assuming the fixed side pays annual money. In this case, the nominal value of each of the fixed flows is

$$\text{Fixed Rate} \times \frac{\text{Act}}{360} \times \text{Notional}.$$

It is assumed that the fourth year in this swap is a leap year and thus accrues for 366 days. Note that the floating side, which still pays 1-year LIBOR annual bond, is unchanged from the swap we modeled in Figure 3.6.

As shown in Figure 3.8, the fixed rate assuming the fixed side pays annual money is 5.392%. Our approximation of 5.395% was only three-tenths of a bp off from the answer provided by our model.

One More Question

What is the present value of \$100mm received at the end of year 4? Well, there are a couple of ways to go about this. We could use the 4-year zero coupon rate and compute the present value as

$$\frac{\$100\text{mm}}{(1.0494)^4} = \$80{,}739{,}908.$$

How else could we solve for this? Remember that on any reset date, the present value of the fictitious floating rate bond is par.[21] In Figure 3.6,

[21] We did say earlier in this chapter that we would show that this is true. To this end, consider a \$100 swap that has T years and pays annual coupons on both sides at times $1, 2, 3, \ldots, T$. Denote by $f(t, t+1)$ the forward rate between time t and $t+1$. Now suppose we are standing at time $T-1$. We know that in one year at time T we will receive the principal of \$100 and the last floating coupon. The forwards indicate that the expected coupon to be received is $f(T-1, T) \times \$100$. Using the forward $f(T-1, T)$, we can discount the cash flows at time T to time $T-1$ as follows:

$$\frac{[\$100 + f(T-1, T) \times \$100]}{1 + f(T-1, T)} = \$100.$$

Now take another step back one year to time $T-2$. In a year from time $T-2$ we expect to receive the floating coupon $f(T-2, T-1) \times \$100$, and we will also have all other cash flows to come, which we found was worth \$100 at time $T-1$. We can discount this back to time $T-2$ using $f(T-2, T-1)$ and find when we are standing at time $T-2$ the value of all remaining cash flows is:

$$\frac{[\$100 + f(T-2, T-1) \times \$100]}{1 + f(T-2, T-1)} = \$100.$$

There is a pattern here. When we are standing at any time t, we know that in a year from now we are expecting to get the coupon $\$100 \times f(t, t+1)$ plus all remaining cash flows in the swap that are worth, as we just showed, \$100 at time $t+1$. Now, going all the way back to time 0, the effective date of the swap, we know that in a year at time 1 we are going to receive the fixed coupon $\$100 \times f(0, 1)$ and all future cash flows in the swap, which have a value as of time 1 of \$100. Discounting back to time 0 gives a value of par.

So what if today is not a LIBOR reset date in the swap? Is the value of the floating side still par? The answer is not necessarily. Let's consider, without loss of generality, the case in which we are currently in the middle of the first year of the swap. The effective date (and first LIBOR reset date) of the swap is time 0, the first payment date will be at time 1, and the current time is denoted by τ, where $0 < \tau < 1$. For example, if the effective date of the swap was nine months ago, then $\tau = 0.75$. Denote by $L(0,1)$ the first LIBOR setting. This is the LIBOR that set at time 0 and will be paid at time 1.

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	5,470,000	5,209,524	5,000,000	4,761,905
2	5.113%	5.226%	5,470,000	4,950,807	5,225,748	4,729,739
3	5.312%	5.712%	5,470,000	4,683,288	5,712,202	4,890,656
4	5.494%	6.041%	105,470,000	85,156,381	106,041,364	85,617,700
				100,000,000		100,000,000
		Fixed rate	5.470%	NPV	0	
		Notional	100,000,000			

Figure 3.9: The Swap Assuming Principal Exchanged

we see that the present value of the floating coupons is \$19,260,092. So this implies that the present value of the principal received at maturity is \$100,000,000 − \$19,260,092 = \$80,739,908.

In order to verify that the present value of the fictitious floating rate bond is indeed par, take a look at Figure 3.9. This is the same swap that we just modeled; however, we are now making the artificial assumption that the two counterparties exchange principal at maturity. Of course, the two counterparties do not literally exchange principal at maturity, but including fictitious principal on each side does not change the value of the swap at all. The swap rate is as we found before. Moreover, note that the present value of the floating side, including principal, is par. In addition, since the swap is an NPV = 0 transaction, this means that the fixed side is also worth par, as we see is the case in Figure 3.9. It's like that song by Radiohead about everything being in its right place.

A very similar argument to the one above can be used to "bring back" all of the outstanding cash flows on the floating side to time 1. The present value as of time τ of all cash flows of the floating rate bond is

$$\frac{[\$100 + L(0,1) \times \$100]}{1 + (1-\tau)L(\tau,1)},$$

where $L(\tau,1)$ is the spot LIBOR rate from time τ to time 1, and the factor $1-\tau$ preceding this term adjusts for the appropriate day count. In general, this expression does not equal par, and thus it is not the case that the value of the fictitious floating rate bond is par on a date that is not a reset date.

Maturity of Swap	PV01 w/ LIBOR Set	PV01 w/o LIBOR Set	PV Coupon 01
1 year	$ 0	$ 9,525	$ 9,524
2 year	9,053	18,577	18,575
3 year	17,617	27,141	27,136
4 year	25,694	35,219	35,210

Table 3.9: PV01s and PV Coupon 01 in a $100mm Swap (Value in Dollars) Given the Swap Curve in Table 3.5

3.2.2 PV01s in Our Stylized Example

In Chapter 2, we introduced the notions of PV01 (both with and without the first LIBOR set) as well as PV Coupon 01. We can apply the same concepts here in our stylized example in order to measure interest rate sensitivity of swaps. Table 3.9 provides these computations.

Note that the relatively large differential between PV01 numbers with and without the LIBOR set is a function of the assumption of 1-year LIBOR in our stylized example. Setting the first LIBOR in our stylized example (which sets a rate for an entire year) has a greater impact in reducing PV01s than in Table 2.3 in which a LIBOR setting is only for three months. In the case of a 1-year swap in which LIBOR has been set, we see that the degenerate nature of our stylized example results in a zero PV01.

3.3 Moving On: Pricing Up Nonstandard Swaps

We have barely gotten off the ground yet. In Section 3.2 we saw that we could regard the swap curve as a par bond curve and exploited this fact to derive zero coupon rates and LIBOR forwards. We then used these zero coupon rates and forwards to model and price up (i.e., solve for the fixed coupon in) a 4-year swap. And we found the following completely unspectacular result: the 4-year swap rate that we solved for in our model was equal to the 4-year swap rate. That is to say, we started out with a swap curve as given on a composite page, and from our swap model we found that the 4-year swap rate was equal to the 4-year swap rate that we saw on the composite page.

As suggested, this exercise was not a waste of our time. The bootstrapping technology that we have just developed is going to be the underpinning of virtually everything else that we price from here on out. Obviously, there was no need for us to model up a simple 4-year swap, since we could observe the 4-year swap rate on the composite page. But sometimes it is nice when working on something new that things turn out the way we would expect them to.

Handchecks

Since the swaps that we are going to look at now are not going to have prices that we observe on a screen, we are going to emphasize, where possible, something called a "handcheck." A handcheck is a quick and dirty way for us to price up a swap on a financial calculator. The handcheck will not be exact and is no replacement for calculating the answer using a model, but it is a good way to check that the answers we get make sense.

We actually already worked through one example of a handcheck. In Section 3.2.1, we modeled up a 4-year swap assuming the fixed side paid annual money instead of annual bond. In Figure 3.8, we found that the fixed rate annual money was 5.392%. But whereas the 4-year swap rate quoted annual bond could be found by looking on the composite page, we could not look there to find the rate quoted annual money. It was therefore comforting when our approximation of 5.395%, as computed in Equation 3.8, came to within a few tenths of a bp of the answer our model gave. Our model may as well be regarded as a random number generator unless we have some way of justifying the answers that it produces.[22] This gave us confirmation that we modeled the trade correctly.

We are still going to rely on the bootstrapping model to price up a wide variety of swaps. But learning to handcheck a trade is an invaluable tool. We want to make sure the answer that our model spits out is in the realm of reality. We should focus as much on the handchecks that we will develop as on the pricing our model gives us. Otherwise, it will be hard to know if we have made a modeling mistake or accidentally suffered from hitting the wrong key on our keyboard. It will be nice to have a way to make sure that the answer we're getting from our model is somewhere in the ballpark.

Let's keep in mind handchecks are only approximations. They are back-of-the-envelope calculations, making simplifying assumptions in order to get quick verification that a model has given us the right number. Not every trade can be handchecked, but for those that can, it's very reassuring when the results of a pricing model agree with a handcheck.

[22] Many in the swap industry are taught to never say "done" on a trade unless either it is possible to justify the number that a model came up with using a handcheck or a colleague independently verifies the pricing.

3.3.1 Mark-to-Markets

In this section we will look at pricing swaps that have been in existence for some time. When a swap is first executed, the fixed rate is such that the swap is an NPV = 0 transaction. Yet as we saw in Chapter 2, as swap rates move around, the value of a swap that was previously executed will usually not remain zero.[23] That is precisely why investors enter into swaps in the first place. An investor who enters into a swap in order to hedge an asset or liability or to take a view on the market is counting on the change in value of the swap, given the passage of time and the change in swap rates. To price an existing swap in this context means to determine its current market value, or *mark-to-market.* Let's work through a few examples now.

Example 1

Referring to our stylized example and the swap curve in Table 3.5, suppose that three years ago a client entered into a $100mm 7-year swap with a dealer in which the client received fixed at 7.00% annual bond and paid 1-year LIBOR annual bond. What is the value of this swap today?

The first thing to consider is whether or not the first three coupon payments on each side of the swap that have come and gone should be considered when valuing the swap today. These coupons have absolutely nothing to do with the value of the remaining four years in this swap.[24] The only thing that matters is valuing the remaining cash flows left in this swap.

For all intents and purposes, this swap is now a 4-year swap. This swap was put on three years ago, and now has four years remaining. Its value is going to depend on where the current 4-year swap rate is in relation to the fixed coupon in this swap: if this swap has a fixed rate higher than current 4-year swap rates, then it will have positive value, or will be *in-the-money*, to the fixed rate receiver. If the swap has a fixed rate lower than the current 4-year swap rate, then it will have negative value, or will be *out-of-the-money*, to the fixed rate receiver. Since the coupon on this swap is 7.00% and the 4-year swap rate is 5.47%, we know that the swap is going to have positive value to the client who is receiving fixed and negative value to the dealer who is paying fixed.

Let's take a look now at valuing this swap with our model. Let's model up this trade from the perspective of the dealer and emphasize who is getting

[23] As we will discuss in Chapter 8, even if swap rates do not move, the value of a swap will in general change with the passage of time.

[24] If we were to value a stock, would we include past dividends when using the dividend discount model of stock valuation? No, we wouldn't. The same principle applies here.

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	(7,000,000)	(6,666,667)	5,000,000	4,761,905
2	5.113%	5.226%	(7,000,000)	(6,335,585)	5,225,748	4,729,739
3	5.312%	5.712%	(7,000,000)	(5,993,239)	5,712,202	4,890,656
4	5.494%	6.041%	(7,000,000)	(5,651,794)	6,041,364	4,877,792
				(24,647,284)		19,260,092
		Fixed rate	7.000%	NPV	(5,387,192)	
		Notional	100,000,000			

Figure 3.10: Marking to Market a 4-Year Swap with a Coupon of 7.00%

what in this swap by signing the cash flows accordingly.[25] As shown in Figure 3.10, the dealer is receiving the LIBOR cash flows that are worth \$19,260,092 in present value terms and is paying out the fixed coupons of 7.00% that are worth \$24,647,284 in present value terms. Thus, the swap is worth \$19,260,092 – \$24,647,284 = −\$5,387,192 to the dealer and is in favor of the client by the same amount.[26] That's a lot of dough for the dealer to

[25] Sometimes, such as in the swap model shown in Figure 3.6, we won't bother to sign the cash flows of a swap. In such cases the NPV of the swap is equal to the difference between the present values of each leg of the swap. In other cases we will want to emphasize who is getting what in the swap and we will sign cash flows accordingly, with positive cash flows representing cash that is received and negative cash flows representing cash that is paid by a particular counterparty. Of course, when we sign cash flows then the NPV of the swap is equal to the sum of the present values of each leg, not the difference between the present values of each leg.

[26] Implicit in these calculations is the assumption that these cash flows should be discounted at LIBOR. Despite the fact that the cost of capital of various financial institutions may differ from LIBOR, this has nonetheless been the market standard for calculating the mark-to-market value of a swap. Since the time of the financial crisis, this assumption has been called into question by many in the industry.

In fact, it would not be an overstatement to say that the industry is going through a (not so quiet) revolution regarding the issue of what rates to use for discounting purposes when computing the mark-to-market of a swap. At the root of the issue is the CSAs that many counterparties have in place with one another. As discussed in Chapter 2, a CSA provides for the transfer of collateral from one counterparty to another on the basis of the mark-to-market value of a swap or portfolio of swaps. Furthermore, for USD swaps, the CSA typically requires that interest on the collateral should be calculated based on the Fed Funds overnight rate.

When the difference between LIBOR and the Fed Funds overnight rate was small and not particularly volatile, most people in the market ignored this issue. But when the be down on the trade. Presumably the dealer hedged himself with Treasuries or Eurodollar futures at the outset of the trade and earned his bid/offer on the trade and was immunized against changes in the value of the swap.[27]

Handcheck for Example 1

The value of the swap that we just modeled is not going to be found on any composite page. A composite page provides current market swap rates but does not give the value of swaps with off-market coupons. It would be comforting if we could come up with a way to make sure the value of the swap that we just computed makes sense.

To this end, note that in the remaining four-year life of the trade, it is as if the counterparty is receiving an annuity of 7.00% − 5.47% = 1.53%. This can be seen in Figure 3.11. Let's work through the "swap math" in this figure in which we've used a little trick. The top box represents the value of the swap in which the client is receiving fixed at 7%. Notice that the value of this trade, whatever it may be, is equal to the value of a swap in which the client is receiving fixed at 7% (i.e., the very same thing) plus the value of a swap in which the client is paying fixed at 5.47%. This might seem kind of weird. In algebraic terms, we just said that $A = A + B$, which is only true when $B = 0$. But this is indeed the case here. The value of a swap in which the client is paying fixed at 5.47% for four years is zero, since 5.47% is the

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LIBOR-OIS spread blew out during the height of the financial crisis (as discussed in Section 1.2.1), people became very aware of this issue. As a result, many in the industry believe it is more appropriate to discount the cash flows in a swap by zero coupon rates derived from an OIS curve, rather than from a swap (i.e., LIBOR) curve. Zero coupon rates formed from an OIS curve will be lower than those formed from the swap curve. The result of this is that the mark-to-market of swaps under OIS discounting will be higher (in absolute value terms) than under LIBOR discounting. See Whittall (2010c) for more details.

This phenomenon has become even more complicated and contentious, as some CSAs allow for collateral to be posted in multiple currencies, creating a "cheapest-to-deliver option" in which a counterparty in a swap will be incented to post the cheapest type of collateral, optimizing among collateral denominated in various currencies. Of course, what may be cheapest to deliver today may not remain so in the future, and a counterparty may optimally choose to post collateral denominated in a different currency at some point in time over the life of the swap. See Sawyer (2011) for more details. The pressure in the industry on this issue is causing certain clearinghouses to change the way they value swaps (see Whittall [2010a, 2010b] for more details). Despite these issues, we will for our purposes always compute discounted cash flows using the swap curve.

[27] As we have discussed in Chapter 1, if the dealer hedged himself with Treasuries, he would still have spread risk over the life of the trade. Thus, he would not be perfectly hedged and would face mark-to-market fluctuations over the life of the trade but would face considerably less risk than if he maintained interest rate sensitivity by not hedging at all.

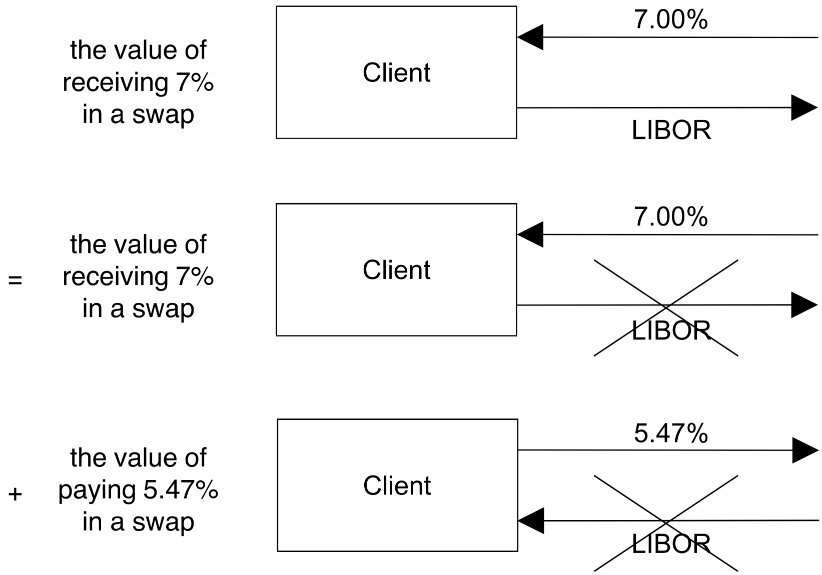

Figure 3.11: Looking at the Value of the 7.00% Swap

4-year swap rate. That is, by definition the 4-year swap rate (which happens to be 5.47%) is the rate such that a 4-year swap is an NPV = 0 transaction, and therefore we can conclude that the bottom swap in the figure is worth zero.

Now looking at the bottom two boxes in Figure 3.11, we can do a lot of canceling out. Note that the client is paying out LIBOR in the middle box and receiving LIBOR in the bottom box. So we can cancel these legs of the two swaps out with one another. That simplifies things and leaves us with the client receiving fixed at 7.00% and simultaneously paying fixed at 5.47%. In other words, the client is receiving an annuity of $7.00\% - 5.47\% = 1.53\%$ of the notional for the next four years.

What is the present value of this annuity? Although the appropriate way to compute the present value of this annuity is to discount each cash flow by its appropriate zero coupon rate, we can come up with a decent approximation using a financial calculator. Let's set

$$\begin{aligned} \text{PMT} &= 1.53\% \\ i &= 5.47\% \\ n &= 4 \end{aligned}$$